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Course Code : CSA15
Course Name : Cloud Computing & Big Data Analytics.

CO4 - AT1

CLASS TEST

1) Unified Big Data Architecture for Retail

Organization :

Problem -

A multi national organization has customer data in different places :

- Online shopping websites
- Mobile applications
- Physical stores
- Social media

Proposed Architecture -

- Data collection : Apache Kafka, Apache Nifi collect data from all channels.
- Storage : Data cake/ HDFS or cloud storage.
- Processing : Apache spark process customer data.
- Analysis : ML analyzes customer.
- Output : Personalized product recommendations.

Benefits -

- Provides real time analytics
- Improves personalized recommendations.

2) Distributed computing for Scientific Research data:

Problem -

Scientific research produces huge amounts of data that are difficult for one computer to process.

Distributed Computing Model -

- Large dataset is divided into small parts.
- Multiplying computers process these parts simultaneously
- Results are combined at end.

Parallel Processing Benefits -

- Faster processing : Many computers work at same time.
- Scalability : More computers can be added when data increases.
- Efficiency : Work load is shared among systems
- Fault tolerlanrace.

Conclusion -

- Distributed computing is more efficient than centralized systems for processing large data scale scientific data.

3.) **Fraud Detection framework from Digital Payments:**

Data Sources -

- Payment transactions
- Location data
- Device information
- Login history
- Customer behaviour.

FrameWork -

1) Collect transaction data continuosly.
2) Use Apache Kafka for streaming data.
3) Use Apache spark streaming.
4) Apply machine learning models to identify suspicious activity.
5) Generate alerts on block high risk transaction.

Benefits -

- Detects fraud in real time.
- Identifies unusual customer behaviour.
- Reduces financial loss.
- Predictive models improve detection accuracy for.

4) Managing Multimedia data in Big Data :

Challenges -

Multimedia data includes :

- Videos
- Images
- Audio files

Problems :

- Large storage requirements.
- High processing requirements.
- Different file formats.
- Slow data transfer.

Recommended Technologies :

- Cloud Object Storage : Store large multimedia files.
- HDFS : Distributed storage for large datasets.
- CDN : Faster delivery for videos and images.
- Compression : Reduces storage requirements.

5:) **Disaster Recovery and Business Continuity:**

A cloud big-data platform should continue operating even during failures.

Strategy -

* Define RTO and RPO.
* Deploy systems access multiple regions zones.
* Use automated filover and load balancing.
* Use specifications and regular backups.
* Monitor systems using tools like prometheus.

Redundacy prevents single points of failure. Retification keeps topics of data in different location.. Automated failure quickly shifts.

6:) **Data Quality and Goverance:**

Poor data quality affects accuracy of big-data analytics.

* Inconsistency produces conflicting results.
* Duplication increases couns ML models.
* Missing values reduce analysis accuracy.
* At large scale, small resource provide major inc.

Governance Strategies :

* Define data-Quality standards such as accuracy, and consistency.
* Use MDM to remove duplicates.
* Standardize schemas.
* Continuously monitor & audit data quality.

7.) ## Recommendation System for Streaming Service :

A streaming source collects user activities such as watching, skipping, rating etc.,

System -

* Kafka - Collects user events.
* HDFS / S3 - Stores raw data.
* Spark - Process data & builds recommendation models.
* Flink streaming - Update recommendation in real time.
* A/B Testing - Measures user engagement.
  User behaviour gives better performance info.
* Serving. Redis/Model - Serving systems provide fast recommendations.

8.) Role of Cloud Computing in Big Data :

Cloud computing provides scalable storage, computing and networking for big-data analytics.

Advantages-

* Scalability - Resources can inc or dec as needed.
* Cost - Pay-as-you-go reduces initial hardware test.
* Flexibility - Provides screwing such as BigQuery, azure etc,
* Managed Operations - Cloud provides handle hardware.

Limitations-

* Unexpected costs from excessive usage.
* Vendor back-in.
* Network dependency and latency.

9.) Real-time Transportation and logistic Analytics :

A logistic system collects GPS, traffic, fuel & vehicle sensor data.

Framework-

* Kafka/MQTT : Collect real-time vehicle data.
* Flink/Spark : Process location & traffic info.

* Storage : Time-series database.

* ML : Predicts traffic and vehicle maintance.

Real-time traffic info allows dynamic route changes, reduce fuel-time & travel-fuel consumption.

10.) Ethical & Logical Issues in Big Data :

Big data collection creates ethical & logical concers-

Ethical Issues-

* Privacy invasion & excessive data collection.
* Lack of informed consent.
* Algorithmic bias and user data.

Logical Issues-

* Data-protection regulations such as GDPR.
* Data-Breach liability.
* copyright issues with third-party data.

Policies-

* Use privacy-by-design and data minimization.
* Obtain clear use consent.
* Conduct fairness audits on ML Models.